higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T460**(E)**(A9)T

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

9 April 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

Copyright reserved

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 A + AB

A 1
B AB
C A
D B

1.1.2 A (A + B + C)

A AB
B A
C BC
D 1

1.1.3 $A + A\overline{B} + \overline{A}B$

A A + B
B A
C 1
D B

1.1.4 $\overline{A}B + AB + \overline{A}\overline{B}$

A $B + \overline{A}$
B A + B
C AB
D A

1.1.5 XY + X + 1

A X
B XY
C 1
D 0

(5 × 1) (5)

1.2 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Bit	A	number of inputs available to be coupled to a specific logic output
1.2.2	Fan in	B	amount of noise on the input of a logic gate without affecting the output logic state
1.2.3	Logic swing	C	single output of a logic gate
1.2.4	Noise immunity	D	current flow in and out of a logic gate
1.2.5	Power dissipation	E	voltage difference between the logic levels of 1 and 0
		F	product of the supply voltage and the supply current

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 An encoder converts decimal to its binary equivalent.

1.3.2 The 4-line to 1-line multiplexer has two select lines.

1.3.3 The decimal system has a base of 8.

1.3.4 Excess 3 code is a nonweighted self-complementing code.

1.3.5 The half adder is used to add three variables.

(5 × 1) (5)

1.4 Convert $56,4_{10}$ to the following BCD codes:

1.4.1 2421

1.4.2 8421

1.4.3 Excess 3

(3 × 3) (9)

[24]

QUESTION 2

Convert each of the following numbers to their binary equivalents and complete the calculation in the binary number system (follow the instruction in brackets):

2.1 $-19{,}75_{10} - 15{,}125_{10}$ (convert the final answer to octal)

2.2 $F{,}C_{16} \div 5{,}4_{8}$ (convert the final answer to decimal)

2.3 $1E{,}7_{16} \times 6{,}5_{10}$ (convert the final answer to octal)

2.4 $1011{,}001_{2} - 1100{,}01_{2}$ (convert the final answer to octal)

(4 × 6) **[24]**

QUESTION 3

Simplify each of the following Boolean expressions:

3.1 $F = A\,\overline{C}\,(B + \overline{C})\,\overline{B}$

3.2 $F = (AB) \oplus B$

(2 × 6) **[12]**

QUESTION 4

4.1 Define each of the following set of counters:

4.1.1 Asynchronous counter

4.1.2 Synchronous counter

(2 × 2) (4)

4.2 Complete the following table with reference to the points on the pulse supplied to a JK master-slave flip-flop:

Position number	Master	Slave
1	4.2.1	open
2	closed	4.2.2
3	open	4.2.3
4	4.2.4	closed
5	4.2.5	open
6	closed	4.2.6

(6)

4.3 Draw the gate equivalent of a clocked JK flip-flop and give the truth table. (10)

[20]

QUESTION 5

5.1 Draw and describe the action of a ramp-type digital-to-analogue converter. (10)

5.2 Draw the circuit of a module-8 down counter and give the truth table. (10)

[20]

TOTAL: 100

Copyright reserved